The Greenhouse Effect

Divya & Camila



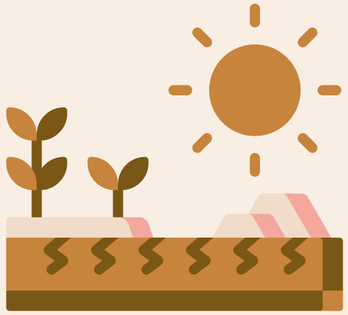
Pick a Side!

Pick which side of the arguement you agree with.



Pick a Side!

Earth’s Temperature without the Greenhouse Effect



Without the Greenhouse Effect, the Earth’s average surface temperature would drop to roughly -18 degrees C (-0.4 degrees F).

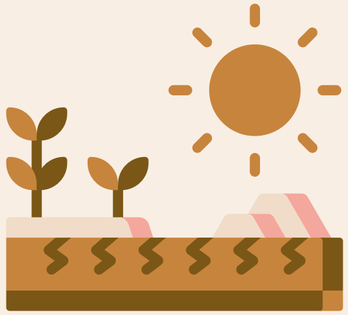


Without the Greenhouse Effect, the Earth’s average surface temperature would remain steady at 15 degrees C (59 degrees F).

https://en.wikipedia.org/wiki/Greenhouse_effect
(calculated by NASA)

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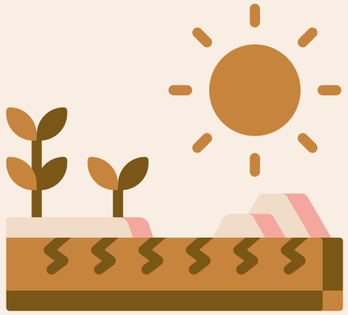


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How do Greenhouse Gases work?



Greenhouse gases work just like a physical glass greenhouse, trapping sunlight inside the atmosphere.

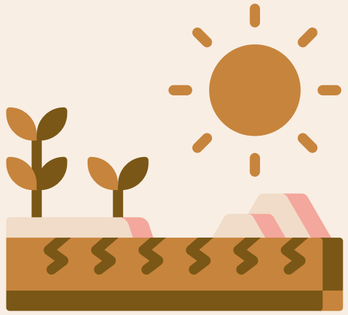


Greenhouse gases do not trap sunlight and instead trap the Earth’s escaping heat.

https://ncse.ngo/misconception-monday-greenhouse-misleading

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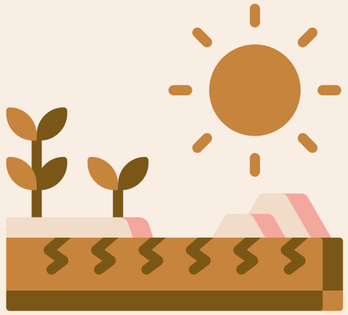


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Which Greenhouse Gas is the most abundant (existing in large amounts) in Earth's atmosphere?





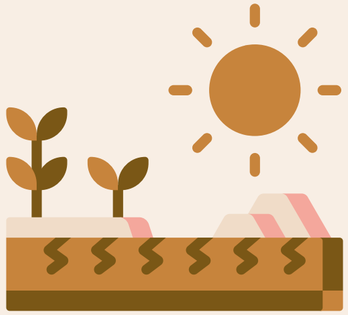
Carbon Dioxide is the most abundant greenhouse gas.

Water Vapor (H20) is the most abundant greenhouse gas.

https://science.nasa.gov/earth/climate-change/steamy-relationships-how-atmospheric-water-vapor-amplifies-earths-greenhouse-effect/

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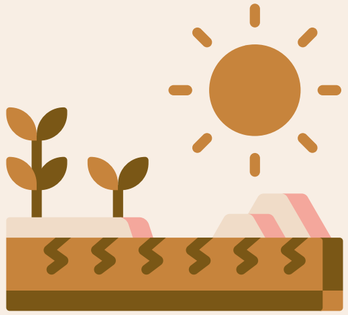
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Why did early industrial societies transition from burning wood (biomass) to coal (fossilized carbon?



Coal has a much higher energy density and a higher carbon-to-hydrogen ratio than wood.

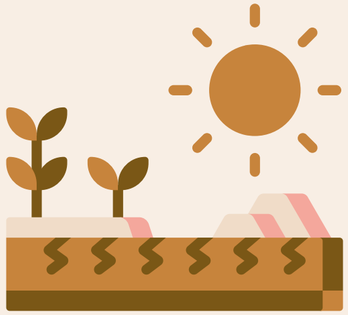


These early industrial societies abandoned wood because regional forests were entirely depleted which made coal the only available alternative fuel source.

https://www.latrobe.edu.au/news/articles/2023/opinion/what-previous-energy-transitions-can-tell-us

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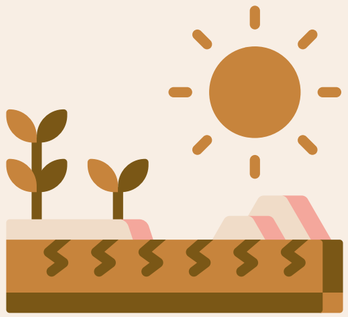


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Pick a Side!

Methane gas is a much more powerful greenhouse gas than CO2, trapping nearly 80 times more heat per molecule over a 20-year period. If so, why does CO2 still pose a much greater long-term threat to global temperature?



Methane is chemically unstable in our atmosphere as it reacts with OH and breaks down, meaning a molecule of methane lasts only 7-12 years. CO2 accumulates steadily over generations.

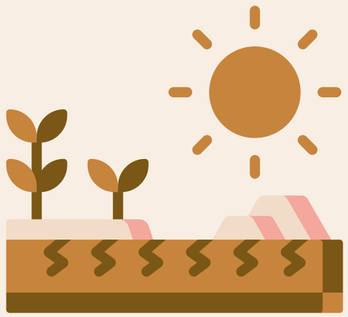


CO2 has a molecular structure that allows it to absorb high-energy solar radiation from the sun. Since methane can only absorb low-energy heat bouncing off the ground, CO2 naturally traps more solar energy over time.

https://science.nasa.gov/earth/explore/earth-indicators/methane/

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Circumstances affecting the Heat of the Sun's Rays

by Eunice Foote



Eunice Foote's Discovery

(2nd article)

What did she discover?

- First paper to experimentally connect CO_2 with global warming
- Used an air-pump, glass cylinders, thermometors, and sunlight to conduct her experiment
- CO_2 traps high heat
  - absorbed more solar heat than regular air
- CO_2 cools down slowly
  - retains the heat
- Damp air is warmer
  - moist air traps more heat
- Dense air holds heat
  - compressed air became hotter

Eunice Foote’s Discovery

(2nd article)

Things to notice:

- Did not actually measure the atmospheric greenhouse effect
- Her setup exposed the glass jars to direct sunlight. The warming she observed was largely driven by the glass itself trapping heat and the gas absorbing direct sunlight, rather than the trapping of infrared radiation (Earth's outgoing heat) that defines the true greenhouse effect.

Greenhouse Effect 101



IMPORTANT ASPECTS

GLOBAL WARMING POTENTIAL (GWP)

Sun heat trapping effectiveness; Measured based on CO_2, usually over 100 years

CONCENTRATION

Usually measured in ppm, ppb or ppt (parts per million, billion and trillion, respectvely

LIFETIME (RESIDENCE TIME)

For how long the gas stays at the atmosphere before being cycled

MAIN GASES

CO2

Main factor in global warming currently; currently at an exponential rise

METHANE (CH4)

Human emotions manly from natural gas management and pecuary

NITROUS OXIDE (N2O)

Long-lasting, high GWP gas; Derived from fertilizers (eg. NPK)

FLUORINATED GASES

Emited in smaller quantities, but have extreme GWP’s and can be highly reactant. (eg. HFCs, CFCs, PFCs, SF6 and NF3)

WATER VAPOR

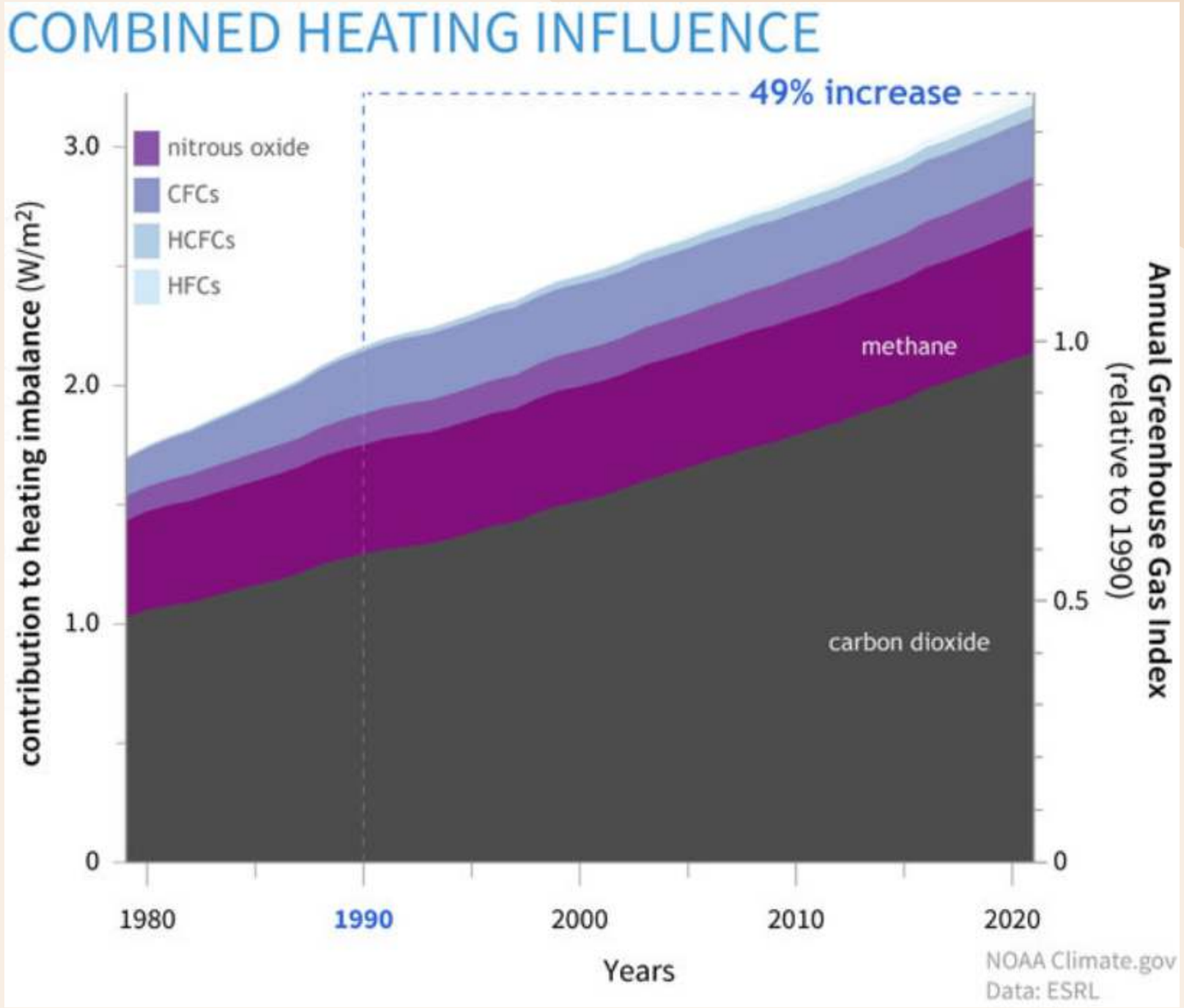
A byproduct of global warming. There are no ways to effectivelly prevent it alone, and has a positive feedback response

TABLE FOR REFERENCE

Greenhouse Gas	Concentration in 2017	Global Warming Potential (over 100 years)	Duration in the atmosphere
Water vapor	Variable with temperature	Less than 1	9 days
Carbon dioxide	407 ppm	1	Highly variable (ranging from years to hundreds of years
Methane	1.85 ppm	25	12 years
Nitrous oxide	.33 ppm	300	114 years
Chlorofluorocarbons	0.0007 ppm	1,600 to 13,000	55 to 500 years

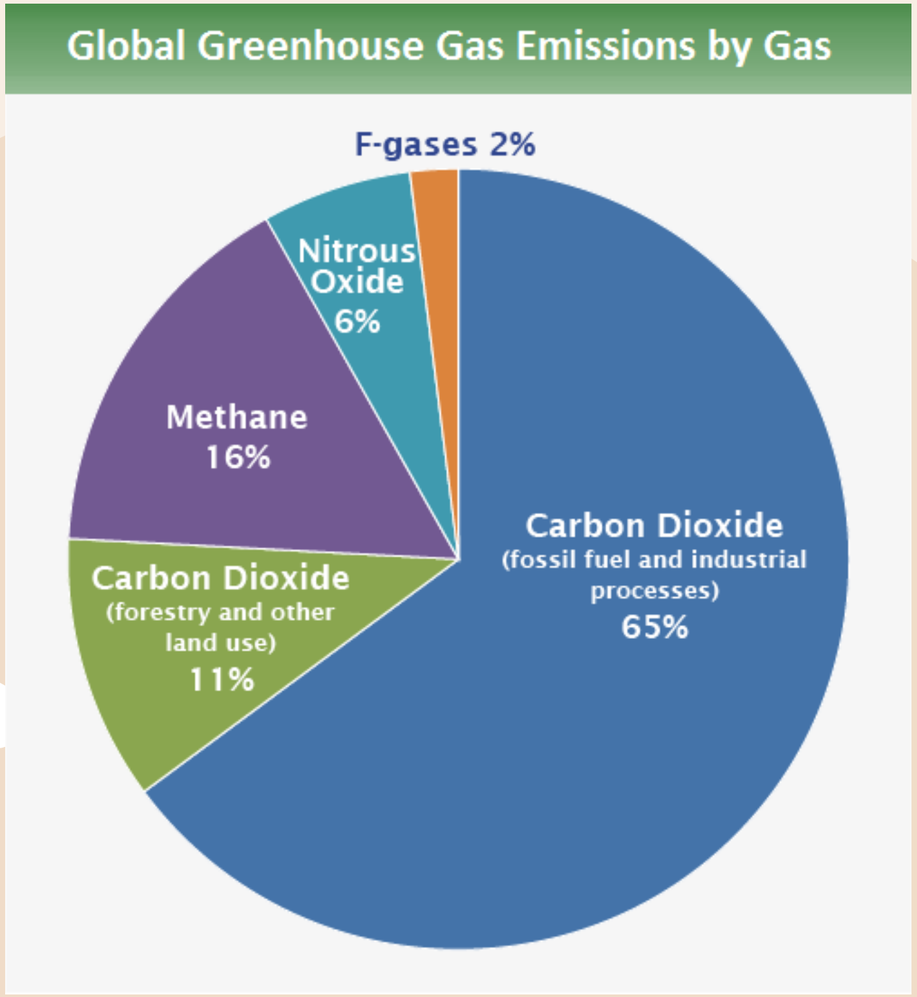
https://mhsapes.weebly.com/day-2-93-95.html

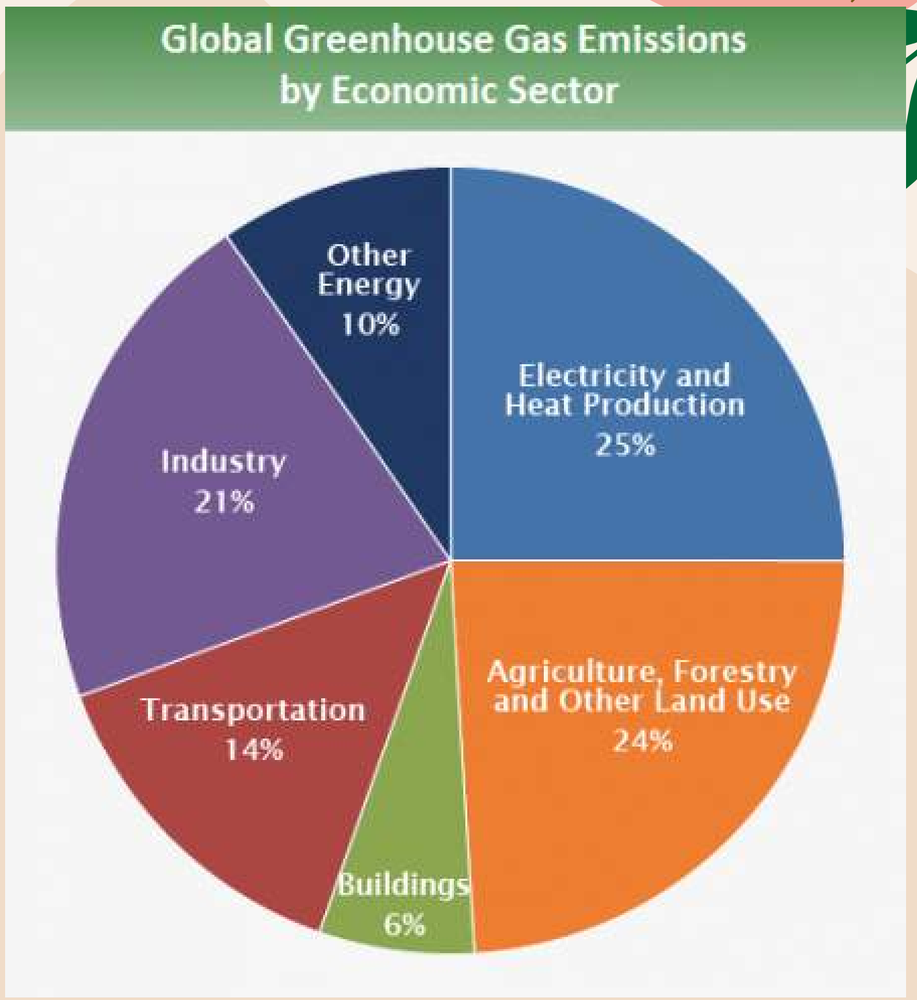
PUTTING IN PERSPECTIVE



https://www.climate.gov/news-features/understanding-climate/climate-change-annual-greenhouse-gas-index

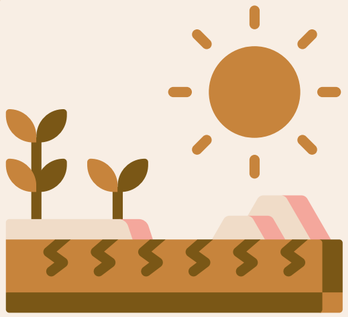
PRODUCTION





https://19january2021snapshot.epa.gov/ghgemissions/global-greenhouse-gas-emissions-data_.html

CONSEQUENCES



- **Intensification of extreme precipitations and weather events (eg. El Niño)**
- **Raising sea levels due to raising ocean temperatues**
- **Alteration of climate systems and ecosystems; Threat to biodiversity**



- **Has more intense effects in vulnerable population and intensifies social inequality**
- **Endangers cities at sea level**
- **May shift populations to urban areas, possibily bringing in diseases**

SOLUTION PERSPECTIVES

STEP 1

GOALS

According to the IPCC, to reach net-zero GHG emissions by 2050, production should be halved by 2030

STEP 2

FOSSIL FUEL REPLACEMENTS

Gradual switch to green energy and investment in energy efficiency is the key element that can mitigate most emissions

STEP 3

AGRICULTURAL CONCERS

Mitigating the emissions from fertilizers, deforestation and food waste is a significant decrease altogether

STEP 4

INDIVIDUAL MEASURES

Reducing GHG emissions on a singular basis, such as using domestic green energy, improoving energy efficient, or lowering consumption



If a solution is created, how might the world react?

The Montreal Treaty of 1987



- A landmark global treaty designed to protect the ozone layer
- Phases out the production and consumption of nearly 100 man-made chemicals known as ozone-depleting substances
- The treaty established a step-wise phase-out schedule with different timetables for developed and developing nations.
- Scientists project that the ozone layer will recover to 1980 baseline levels by around 2040 globally, with polar regions recovering slightly later

https://www.unep.org/ozonaction/who-we-are/about-montreal-protocol

Why does this matter?

- The **first** and **only** treaty in the history of the United Nations to achieve universal ratification, being signed by 197 nations.
- If such an impactful move was created and has real-world implications, could it happen again?



Thank you!



